

ActInf Livestream #022.2 ~ Psychophysical identity and free energy

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okay hello everyone we are live welcome to actin flab live stream number 22.2 nice repeating digits there we're here with alex kiefer and others we're going to be having our second follow-up discussion on the paper we've been discussing for the little last bit psychophysical identity and free energy it's may 25th 2021 and this will be a great discussion so thanks again for everybody who's joining live as well as watching in replay if you're watching live especially feel welcome to add a comment in the live chat that we can address welcome to actin flab everyone we're a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links that are listed here this is recorded in an archived live stream so please provide us feedback so that we all can improve our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams we are here in the second of the two discussions in late may 21 and happy to have alex joining us for both of these discussions which has been great today in 22.2 quick correction we will be just trying to learn and discuss enjoying these conversations seeing where they go enjoying what questions bring up and also carrying on with a few of the topics that we knew that we wanted to cover from last week and with that being said we can jump right in so we can go to the introductions and we'll just um maybe go around introduce ourselves say hello and then end with alex and alex would be happy to hear your thoughts or reflections in this past week so i'm daniel i'm a postdoctoral researcher in california and i'll pass to dean hi folks i'm dean i'm from calgary and uh yeah i'll pass it to adam uh i'm adam i'm a postdoctoral at the johns hopkins center for uh psychedelic and consciousness science and stephen oh hello i mean stephen i'm in toronto i'm a community development practitioner and applied theatre artist and i'm doing a practice based phd at the moment and i will pass it over to alex i think you're the only alex right now okay okay i wasn't sure there's a fellow jester i watch it um oh so i'm alex kiefer um thanks for having me back again to talk about this paper um i'm a philosopher i'm currently affiliated with monash university and i'm also doing more

applied work for nested minds solutions uh it's a kind of active inference ai startup cool um there's a few ways we can start but why don't you feel free to give any opening yes and then we can ask questions we have some other slots right so you asked about my reflections over the past week um i haven't reflected a whole lot on deep themes um i did uh go through so in the point zero discussion there was a um sort of a question about how we get from two point one equation two two one to two two um with the flipping of the uh p's and q's so that strikes me as like not the most central theme we could spend time on but i did uh just spend some time writing out the derivation if that's something we can get to at some point um but in general like i guess my feeling is i think last week was really fun and we kind of it was very far-ranging um i i think it would be fun to like to drill down a bit more on some of the specifics this week but you know it's kind of also whatever people want to say so that's that's what i'd be interested in doing thank you alex welcome blue if you'd like to just say hi we're having intros and then we'll sort of walk through some jokes and questions good morning i'm blue knight i'm an independent research consultant based out of new mexico cool so there's a bunch of places that we wrote down to jump in um maybe actually we could go to dean's three functions walk into a bar so it's slide 11 and i'll have it up large on the stream so dean why don't you unpack what you were thinking here and let's get some initial reads to get the contrastive ball rolling well essentially what the the point two aspect of this process does is it takes um [Music] basically what we've introduced as as as the topic uh tried to gain some interpretation around it and then so what does that mean what does that turn into and so um one of the products of that for me was uh i kind of went back and reviewed the um the 22.1 and one of the things about what you mentioned alex was the explication of a simple idea and i thought that was really important and so what can that what can that turn into that simple idea that was the the three functions walk into a bar piece and essentially spoke to what scott was talking about last time in terms of paradoxes and what do those mean and when do we model and when do we follow but i thought the bottom part was was an important piece in terms of you mentioning uh that you saw continuities and i wondered if you were referring to cynicism and that idea of how we take discontinuities and turn them into continuities man that's that's really interesting so i had never heard of cynicism i keep coming to pier paris through like strange avenues that were not planned um i never consciously thought of it this way but this kind of nicely summarizes like how i see things and and the the way i've just been sort of forced by experience to approach philosophy it like like like i i found it difficult or impossible to draw to draw boundaries that would create discontinuities between categories quite

generally so you know that gets into things like the sororities paradox there's plenty of examples of this but i think it's it for me it's much harder to find a discontinuity than to uh yeah than the opposite okay can you maybe just walk us through these three functions walking into a bar what what is the narrative here and where does it sit in relation to the paper you're asking me yeah yeah okay so so essentially if we're if we're gonna if we're gonna talk about contrastive divergence which i think is a critical path to understanding how we move down a gradient and arrive at some sort of um generative model i think that there are three functions that essentially exist in that it's an inversion process which i again i'm not going to go into the the the reads on that there's an identity piece which is spoken to explicitly in in alex's and then there's the contrastive diverse contrast of divergence piece which enables a person to remain balanced despite the fact that they don't have a particular answer that there are hidden states and i think what a lot of what was being spoken to last time uh scott mentioned some things blue uh stephen was things like scale which don't necessarily we don't come at it with with the same perception as to what we're looking at some of us look at things a little bit on a grander model and some of us look at an on a granular model and i think that the part of being able to fit that into a a form like the joke the three three guys walk into a bar or whatever kind of speaks to the fact that um if you think that this is going to be about stability you're going to get interrupted by the change aspect of it if you think it's going to be all about the change piece you're not going to be able to realize any sort of conceptualizations because conceptualizations require stability so you have to kind of be able to play in both fields at the same time and that's hard and i think that's why alex's paper probably knocked me down about five times when i first read it because you actually did alex try to somehow pull together the ideal idea of stabilizing something even though things were pretty choppy and and rather dynamic so i don't know if this is an example of where the part two of us discussing your paper goes but i think it's it's interesting especially in the context of cynicism and your idea that we're constantly trying to move from the unknown to the known as opposed to the traditional this is what i know and i'm moving into these unknown spaces that reminds me of then anyone alex or anyone else reminds me of sort of two complimentary ways of thinking about education with the tabula rasa the blank slate that inscription is put into it's like filling the cup with knowledge or inscribing into something that's templateless and then the alternative is to burn away ignorance and so it's almost like a moot whether we're moving from the known into the unknown or the unknown towards the known they're they're are left in our right hands of education right yeah yeah this is really cool i mean it it seems like um uh dean like you have

you have an act for like reading reading things in a sort of very uh deeper almost allegorical way we're like i'm talking about it yeah yeah i read charles saunders pierce two decades ago before it became popular right so yeah so so i think it's interesting and it's it's all um these connections are really there but um i just just to to comment on this for a second um so so so i do think that there's something about just the how ubiquitous sort of dynamism is and all this and how there's in a sense there's no stable process that we can latch onto here but when i was talking about contrast of divergence my main point was just um well look that's i'm thinking of that very narrowly as a machine learning like an unsupervised learning algorithm right which i i know you have that in mind too but you're also taking a broader picture um so um my point in contrasting contrastive diversions to uh simulated annealing was uh we can suppose that the brain is running something like contrastive divergence where it's trying to it's trying to reach a lower energy state um but you know the way that algorithm works of course you don't need to actually run right the basic idea um is that you don't need to so so okay sorry let me take a step back for a second um right so like one way of doing this sort of unsupervised learning would be to to take the distribution over hidden states in a neural net that's induced by an input and uh try to uh lower the energy of those states while raising the energy of the fantasies that the network would produce top down right um and so the ideal way to do that would be to run run the network until it reaches its equilibrium distribution and sample from that when you're doing the negative the negative phase of this of um increasing the energy of the or the therefore lowering the probability of the fantasies um contrasted divergence just takes a shortcut and just runs this for like one step or two steps or whatever um so i just want to make that clear it i don't know if it's clear for everyone in the audience who hasn't looked at this deeply um that that's the the specific contrast that i wanted to draw there was just was just between that and a process like simulated annealing where if you if you run it to the to the end of the process you end up reaching the lowest possible energy state um so the brain i guess my point is the brain doesn't necessarily need to be settling to and this does i think connect directly to your point it doesn't need to be settling to a completely you know stable state or even um uh any particular like metastable state it just has to be there has to be this gradient right that is available um but i do think that there's a deeper point here right so the nests or the non-equilibrium steady state that um organisms are are approaching uh as they as they do this variational inference thing is not any kind of final um you know it's it's not it's not a static state like some kind of uh ultimate equilibrium right well going down a set of stairs the one the one steady thing is that i remain balanced and not go tumbling to my eyes right that's the one

steady thing the one balance thing and then there's the gradient itself that's why the there's a next slide that kind of speaks to the that piece of it and i that's all that's all i basically wanted to bring up because i think and i think the cynicism piece embraces or entails all of that there's a there's a dynamic piece and there's a balancing piece and you and you want to be able to sort of address both that's that's cool yep thank you dean stephen and then adam there's interesting questions here for pragmatism and social constructivism you know in terms of how these ideas roll out into how we might talk to other people in the world around projects and stuff because of this implication so i i wonder how you you would bring in this kind of embodiment of that so um if the brain is an organ that is doing expected more of an expected free energy with the nervous system to help to manage the movement of the organism and the organism overall has got that variational free energy to maintain homeostasis you you you've got this you've got this energy balance piece but you've also got that need for entropy so i wonder how with how this can inform us just like people when we do i don't know if you know about the alexander technique but it's when someone's when you think about the body and the skeleton we think about it as just the bones but actually it's really everything's held together in a muscular sort of watery yeah and and and with the tendons and everything so you there needs to be some noise in there that that entropy piece needs to be there when you're going down the stairs and that you can't be frozen you're full yet you need to have some of that energy equilibrium because again you're full so there's i wonder how if we were to talk to someone out there about their life and their world um walking down the stairs what what would we translate this in their language alex if you want to give a thought on that i'm thinking about it um i'm not sure beyond saying that i do think that there's a balance of two sort of uh forces or quantities here always um but i don't this reminds me actually i'm not gonna try to put words in your mouth but this reminds me of uh adam of your your remarks about free will in relation to degrees of freedom and entropy and stuff um so that's a segue but it doesn't have to be taken up cool yeah it reminds me of stream of consciousness or walking downstairs you can't just stop there statically can't stop anywhere along the staircase and you can't halt the stream of thought so what is happening that's allowing time t plus one to be generated from time t the state only has to move by one moment at a time it doesn't have to do the ultimate um steady state to the lowest possible energy so adam and then anyone else raising their hand um i reread your uh paper last night and it's even better the second time um i'd like to actually talk to you later about like getting into the weeds about different um uh like mappings on the neural level of like between the thermodynamic and the informational um but yeah before we move on i was

wondering how you'd feel about um like a multi-phase uh or a multi-part description um of neural architecture where like there's some aspects which might be more annealing like in some aspects that might be more contrastive like so like you could like for instance tell a story of like some sort of um uh predictive workspace where like you can think of it as like that the model selection is like iterative annealing potentially in terms of some of its aspects um but then like it's just like around like you you could think of like like like the hain style ignition event as you're generating this this big high temperature complex that then settles down and selects the model but this would just be like one of the things that's happening like something contrastive could be like the thalamus like maybe like orchestrating different cycles and comparing them more or like the hippocampal system and that could be like doing your contrast like would that be copacetic or with that right yeah no that sounds that sounds awesome i mean in general it seems consistent with like the way i've just i've learned to see these things you know just just paying attention to all these different sources of evidence it seems like um yeah this kind of um it's it sounds right to me i guess my only uncertainty is like the reason i focused on um on this contrast was in part i wanted to argue for as you know for an identity between the the energy free energy in the thermodynamic sense and in the variational sense relevant to variational inference uh i needed to i needed to make explicit the argument that whatever's going on even if if there's sort of a um multiple time scales in which this is happening or or if there are different parts of the system that are um in effect being annealed to different degrees uh that the whole thing it's not possible to think that the whole thing settles to like you know the lowest possible energy state while it's alive so anyway i don't mean to like deflect i think what you just brought up is really interesting and you probably know more about the relevant you know neuronal mechanisms that i do so i can't critique that idea but it sounds as he said copacetic yeah adam you want to add something there oh i mean yeah and i wouldn't want to like distract him like the point of if we you actually you have to not be an equilibrium you have to have this like open endedness and dynamics to keep the process going um but i and i i don't know like how productive like sometimes like i'll squint it like different machine learning algorithms and like you know push them a little bit further so like you could think of this entered a bayesian model selection it's like a kind of annealing but that could like not be productive potentially even though aspects of it could be described that way yeah well that's a good i mean it's kind of a good way of framing it like i think this goes back to this synchrotis thing like you i think you can probably always redescribe something in terms of something else but it's a question of how useful it is and um and i never exactly know

where the boundary is right is what becomes just kind of a fanciful like philosopher's trick versus what's useful but the thing that you're suggesting sounds like it would be a useful way of thinking about things and and when i contrast contrast of divergence and annealing when i think of annealing i think of dna in the test tube as you turn the temperature down so you come to a final frozen state that's like a fixed crystal essentially whereas the idea of contrast of divergence it's more like cybernetics it's like a guiding in a navigation on a flow already and so even if you didn't have any divergence at all between your north star and your header where you're going and where you want to go you still would be moving and so in that sense even if you're trying to minimize the divergence there still is a sense of movement whereas in simulated annealing the idea that there's no difference is more associated with a crystalline or a static state so they have very different sort of telios almost and for dynamic systems that person going down the stairs stream of consciousness action selection in an uncertain world that feels more like the cybernetic navigation rather than the dna in a test tube though those are just some aspects so stephen and then adam that that idea of um how things keep going and the dynamics um i'd be interested in your thoughts about how that might play out in in behavior or in the world so i could imagine in the evening being a little bit like if things are trying to hibernate there may be a state where they can go closer to like like glass glass gets annealed basically that's how you plunge into water and it it holds that liquid state even though it's colder than liquid but the um the question i have is is how do you reconcile some of these um these these these questions around where we can get caught in loops for instance you know that um casper hesp talks about depression where you're seeing a lack of affordance in the environment and that could be confirming your predictions you can get kind of caught in a loop and some of those challenges in computational psychiatry are they're sort of low energy i actually haven't got an answer it's made me think but they're sort of low energy states but also like looping conformationally states you know where you get a reduction in free energy relative to some some sort of stick in the sand that has been established for the priors and the dynamics and then it kind of gets stuck in that you know be it you know hyperactivity something that's outside of where the organism want to be but it's still going there and i know if that's quite the same as um going into the sort of a low energy equilibrium you know going to it's not necessarily going down in energy it's sort of getting caught in a loop though of variational free energy yeah that's that's really interesting actually um sorry it was someone else wanted to oh no alex you're going to hang up and then after your thoughts will go to adam yeah so i mean my my my intuitive reaction is that is that yeah something like that happens in

depression there's a sense in which it's a lower energy state um in my my paper which is kind of like a sophisticated argument for a really simple-minded idea i want to identify those two things and say yeah it's physically a low-energy state as well um but it's a local one so uh again this is my intuition at the moment i don't know if it'll be borne out or if it'll work but it seems like sometimes in depression maybe we reach like a yeah like a local energy minimum in a certain part of the network or something but it's not part of an overall optimal state for the organism when it comes to behavior and things like that uh it's really interesting to think about how that happens yep thanks adam um uh actually to speak what uh steven just mentioned um there's a um within like the psychedelic uh research communities there's uh interesting models based on um therapeutic change through annealing where mike johnson at the quality research institute has something called like the neural theory or theory of annealing where it's the psychedelic state increases the temperature helps you to break out and then you can crystallize in your regime and uh robin hart harris is described in very similar terms um one uh thing about like generalizations of annealing uh that came to mind um when dan was talking was um uh doug hoff setter and melanie mitchell um with their copycat architecture um they said it's different from annealing but they had something that was kind of like ant colony optimization and kneeling where there would be a temperature parameter for when your analogies weren't fitting and so you can think of like some sort of contrast of bits saying like how good is your like matches how good is like these analogical matching like discrete comparisons and then this parameterizes the temperature of a continuously adjusted annealing process that doesn't settle and and more recently really interestingly is um um well from the beginning the language was completely like work spaced up and uh more recently uh melanie mitchell has like explicitly drawn a connection to the global neural workspace theory and so there could be like a generalized annealing that provides that it has a contrast a bit and an annealing generalized annealing bit that does give you like two kinds of like psychophysical identity mappings maybe and a total a total speculative note would be sometimes we have you know accuracy minus complexity or pragmatic versus epistemic gain maybe some decision making approaches are able to hold both together and maybe even rebalance between those different modes dean i'd like to actually go to 12 and hear a little bit more on what you thought about contrast of diversions before we head to the next topic so what did you see or find interesting on slide 12 what brought you to that slide well i'd actually looked at stuff from oliver woodford before and so that was kind of why again i was i was sort of i was connecting dots in my own head and it's interesting because the parts that adam and steven just brought up in part

in part of our conversations off offline one of the questions i asked daniel and blue was around the idea so you have the physical math and you have the statistical math and somehow they come together and make babies and that's what we kind of have here and so what is this what is this project and all i liked about the not to not to discount or put aside the simulated annealing piece but to give what the what i think happens as a for example when the person goes down goes down the stairs under control as opposed to falling to their death you know lowest energy state i thought that this essentially spoke to it by just giving a nice metaphor in terms of shining the the torch beam in our chosen direction of travel which allows us to see the lowest point in the field in that direction which i think alex was again addressing i don't think alex you meant to make this such a a central point of your paper but i i thought in terms of um trying to explicate the simple idea the fact that you didn't just focus on the fact that there's a gradient but how does a person move down that gradient and not lose control was kind of what i wanted to bring forward here and in the notes at the bottom which i can't see on my screen there's a really nice explanation for what contrast divergence does in terms of being an assistant that's why i want to bring that one forward cool that and then stephen i don't have much of a thought i mean i think it's it's a really cool um angle that you have on this and it's i again it's it's something that's there even if it wasn't my intended focus and it reinforces i think some of the themes that i was focusing on thanks stephen so what i think is quite useful and i'm kind of be sort of bringing this idea if i was trying to explain this to an arts practitioner or someone in it does you know psychotherapy or something how to explain the implications and one thing i think is important here when you talk about that energy minima if i jump to the bottom of a hill i metaphysically i might know that i'm at a lower energy potential energy overall but actually i probably get a higher energy when i hit the floor for a while right as i start to get stressed and perspire and assuming i'm not just completely squished to oblivion so you're in this interesting you know i suppose we do that on roller coasters a little bit but we we there is there's what we are inferring about our states isn't what it is because we don't directly experience being at the lower energy we experience what our sensorial states enable us to infer about the results of our actions so and then from that we compare that to our expectations based on the stick in the sand or the big sticks in the sand which is our phenotype which is what we can then use as something to anchor against so i'm wondering that that has an implication about how people think about the world because of these types of ideas you know there's there's a sense that they are connected and interconnected at the same time they're always separate in terms of being able to touch something because they're not touching

something they're getting sensory data which makes them get that allows their model to predict what they've just done but they didn't actually touch in the way that people might think they've touched so i'm just putting that out there as some question yep alex go for it yeah that so there's there's this piece um of what you keep doing stephen you keep trying to make this like put like put all this in touch with like the body in the world and stuff and i'm like no let me just stay in my internal model and not worry about that but so i don't know actually i don't know if i if i understood the thrust of just like the last couple of sentences but i think the point that there's the fact is we don't we're not yeah we don't just want to like you know leap to the lowest energy state possible um become what may right it's it's much more about the process and um i think what you said about the this thing being anchored in the phenotype and you know the priors in general is is is really important um one one another paper that i wanted to write or i don't know if it's a paper maybe it's just a blog post or a tweet or something but um it's just on the fact that uh yeah so like you always have to bear in mind when you're talking about this energy minimization process that this is all relative to a phenotype which itself embodies some amount of energy right and so in fact you might even say that pheno the stable phenotypes are some kind of like optimal tradeoff between accuracy and complexity right just in themselves um so there's this force that's sort of pushing up from below which is just the having a phenotype complex enough to predict anything and you have to work within that like those boundaries so i don't know if that really addresses your point because again i totally missed the embodiment part just like flew by me but it's it's nice just one point and then adam so i really like that example of a person or a body in motion going down the stairs and the difference between the controlled walking down the stairs um or maybe even using an assisted mobility device and then falling down the stairs and both of them can't stop at one moment whether you're tumbling or whether you're walking you still can't stop there so it's always dynamical and you are going downhill and you are converting your potential energy into you know going to a lower energy state so those two are good ways to think about where does bodily control and motor coordination which we'll talk about in 23 soon with embodied skillful performance but let's think about a mountain climber that's somebody who's actually contrastively diverging uphill and so that shows us that it's not just like um the dye is cast and then whether you fall or control on the way down but there's a way in which going to the top of the mountain is like going downhill or annealing or reducing the contrast and divergence relative to your expectations of where you want to be and now you can imagine if the water level is rising going uphill is the survival strategy so we want to have a way of describing

where does control and bodily control come into the picture and then even generalizing beyond just bodies with mass falling in a controlled or uncontrolled fashion towards the kind of cybernetic planning that could actually result in a mountain climber for example going a little downhill and little to the side and then making their way up the mountain so alex then adam then stephen yeah yeah now that's really it's a really good point to bring up especially in this context of you know of the thrust of trying to simplify things by identifying physical and you know psychological things like no sometimes you want to climb the mountain so i mean my my my immediate way of trying to summarize how that would work is just well you've got a strong prediction um that yeah that you want to be at the top of the mountain right that's your sort of that's your sort of desired distribution um it defines your desired distribution over observations and so that's gonna it's gonna drive you to climb the mountain but i think that the way it does that is proximally is that there's going to be an actual i would say an actual physical uh disequilibrium in the brain right that's caused by the discrepancy between your predicted state and the state that you're inferring from your observations and lowering that uh you know that potential uh is what drives the action of actually climbing the mountain so um yeah cool awesome adam and then stephen head's spinning a little bit but hopefully this works because like up down um but i'm wondering in terms of uh something like uh depression um uh could we think of it as like a very high local minima and like if we thought of like what we'd expect so i guess okay what is it that's being predicted at which level of the degenerative model um like what is it they're being is what's what's being predicted like and actively like between the original niche and what's being predicted over the you know whole and like skin encapsulated organism the brain and then subsystems there could be some ultimate they have to come to some kind of harmony but they they could diverge but um the idea would be like thinking of um uh a landscape that like when the energy's high it might be more jagged and you might be more likely to get stuck at like uh local you're stuck at this local minima and you want to get to the the a really low one and so you could think of something like uh niche construction as like looking for a catalyst to like try to like smooth out like make a more navigable landscape or you could think of like the psychedelic would be like in turning the heat so i'm thinking of the difference between like a a thermodynamic versus a kinetically favorable chemical reaction so like you might like yeah it's theory if you get from there there you come up ahead with the anthem with the enthalpy but you can't get there it's not kinetically favorable and so like the different things we're doing is trying to like uh find ways of getting kinetic favorability for thing for this thermodynamic favorability and bring those

into alignment very nice atom you can build that bridge so you don't have to ford the river or swim across or you can just pray for the quantum tunnel to appear so i do stephen and then dean wow okay there's something very quite important here i think because okay we're minimizing this energy gradient okay that's fair enough particularly for robots which are using reinforcement approaches and they're basically working on energy gradients and then you've got the idea okay you can work with approaches i more like this gibbs free energy where the reaction can slightly have entropy but the big key thing as well if you're going to take your agent is that where at the level of the chemistry is the chemistry is all non-equilibrium chemistry so the whole of the gibbs free energy is built on equilibrium chemistry here we've got a non-equilibrium chemistry when you get to the level of the organisms and cells and everything so i think what's an interesting point here is so you're going down the stairs you're trying to fall over you're being a rock climber so overall there's this energy piece around how the agent moves but for the agent to move unlike a robot where you might have little pistons the actual cells and that are operating at this non-equilibrium state and quite how different actually this gap between these attractor basins versus energy basins which we normally think of i actually don't know i don't know i'm interested to ask carl fristan that like what is how how much does it shift things to have and basins are to do with the tractors as opposed to just basins to do with energy like equilibrium energy gradients um and and that's the thing that's different in terms of what's really going on when i'm climbing a mountain or is all my muscles and that at the actual level of making things happen a swarming and trying to enable this kind of biological process to happen and okay from a system perspective i can see the energy being less for me falling on the ground but for my muscles to move and everything um they're they're operating at this non-equilibrium process and which will have energy impacts but also as some other types of dynamics which you don't have you have this non-equilibrium steady state dynamic which you don't get in traditional chemistry so there's something there i just thought i'm not saying i expect an answer from that but i think it's a it's an important point that seems to be a gap it just it really shun out for me there nice thanks steven it's kind of like if it's a ball at the bottom of a bowl then it's it's that's the annealing it's the crystal state with the ball and the bowl but then if you have a neuron that's holding a resting voltage that is being actively maintained with an enormous expenditure of energy but it's its own kind of bowl that it's converging to whether it's above or below its set point it's still doing a type of convergence but it's a dynamically held and you're absolutely right that sometimes the lower level entity or the entity which the larger one is composed of they have their own active dynamics even when

the top level system is at rest or appears to be unmoving so dean and then we have a few other fun topics that we'll get to well i'm gonna see if it's okay to ask alex to jump to his third rail at this point because i'm really really curious about the modified ramsey sentence uh aspect of his paper and what i was hoping he might be able to speak to a little bit about is not just what that what the ramsey sentence is talking about but the implications in terms of different time intervals and what that role that plays in us being able to understand how the modified ramsey sentence works in your paper perfect i switched to that slide um 44. so yeah i was definitely curious about that as well alex what does the ramsey sentence or this entire area of formal logic how did that come into play yeah so right i kind of trumped up the ramsey sentence thing in the last episode but um it's kind of silly of me because there's not a whole lot to it in a way but then maybe that's just how it seems to me from the perspective of where i'm coming from but the reason this is in here is that the whole identity the philosophical mind brain identity piece of this was based on i was mostly inspired by david lewis's approach to that topic and so there there are a couple of influential approaches approaches there's lewis's and there's um uh smart i guess the two big papers um but basically uh what but this piece of formal logic is doing is just trying to capture what um what a theory says about the entities that it sort of implicitly defines or how you could get from um this isn't all explicitly spelled out graphically but how you could get from just a collection of statements that you might have sent to about whatever the topic happens to be in this case we're talking about mental states from that to a uh a theory that's um expressed in general terms and that again implicitly defines the uh the entities that figure as the values of these variables here um so the right so i mean we could briefly talk about this i guess is it useful to go over this yeah you guys yeah so um read out the um or translate it or provide an example sure um so right so i guess i'll start with just the ramsey sentence so we can talk about modified versions in a second but um so the first symbol here as you guys discussed in point zero is just an existential quantifier it says there's some x such that whatever follows uh the second symbol is a universal quantifier over the variable y so it says there's some x such that for all y blah blah blah and here t the predicate is just literally like so say that you had a collection of platitudes or they don't i don't want to complicate things just a collection of statements like uh um when i see uh when i see a cat i'm happy uh when i think about happiness uh i tend to think about sunshine things like that right just some examples um and so you just can join all those just put an and between all of the sentences of that type that you could think of in this domain in this case it's folk psychology um so then you'd have you'd have a a uh a long sentence right it's a giant

conjunction now uh that has some terms that refer to mental states like uh believe arguably see although maybe some people interpret that as a physical state there's definitely blurry blurriness here at the edges but in any case things like believe desire think these are clearly mental state terms right so you'd have this long sentence that has those terms and also other terms and so essentially you just you replace each of the the terms in question the mental state terms with a variable and what's left over is the predicate that defines what the sentence is saying about those the values of those variables right and then you just append these quantifiers to the beginning um just to sort of spell out um what your what kind of claim you're making um about the states in question namely there is such a there is something such that it relates to all other things in this way um and so that's that's the basic ramsey sends it's just a um it's an abstraction that allows you to um to spell out in formal logic and first order practical logic uh like what a collection of sentences says about um and the and the last piece here is the so there's the if and only if the three bars um that's the resellian uniqueness condition so this goes back to russell's theory of definite descriptions so like i got i mean my start by the way was in sort of of language and stuff so that's why this is in here i came to the all this sort of statistics and physics space stuff much later um so so russell you know had this this theory of definite descriptions uh that he was trying to explain how you could have a meaningful expression uh like you know the present king of france um even though there is no such thing so how does how do failures of reference how do we philosophically handle failures of reference and things like um and so his his suggestion was that um you can think of definite descriptions like the present king of france as uh just saying well there is something such that it is a present king of france and there's only one of those things right so it's like a uniquely satisfied description and so that's what this end part does here it says this predicate t which encompasses all of the implicit folk knowledge about mental states in this case uh holds uh holds of something only if that thing is equal to x and so one last piece that's important here is that i took x the the bold letters here x and y to be like uh vectors essentially right so like these are we're talking about many different mental states in parallel we do this uh but if you want to you can just think of it simply as being about one mental state but then you generalize to all of them so i don't know where how clear is that what what what it's it's let me try a uh a little bit of a physical example it's like there's the stream of consciousness the stream of statements what you said with an ant i saw a cat or photons hit my eye and i perceived a cat so one of those is more of a physicalist claim and one of those is more on a mental side and it's almost like we're going to draw a line around the mental claims just so that they can be

definitely interfaced into a formal logic even if the contents of that mental experience might not even have a reference in the real world or it might not be something that exists in the same way that those photons hitting the eyes exist but it allows us to interface with those kinds of statements by agents in a formal logical framework well i mean i guess one part i'd push back on a bit is i mean this sentence here clearly states that this x exists whatever you take that to mean right so it could be that there's different criteria for existence right in different domains or something but it definitely said that that there exists something such that it's related to all the other things of this kind and in the ways that are specified in the predicate so like yeah i think i think the the the main i guess the main point here i so i focused on the ramsey sentence because people were interested in it but like the main point is just um that you can take all the things that people say about mental states as implicitly defining them in a sort of theory right and that that's uh that's um and so lewis used this explicitly in the context of trying to identify mental states with brain states right so just to i'll just say a little bit more about this piece and then i'll shut up for a bit but like um if you have this network of relations that's described by a sentence like this uh you can take that as implicitly defining the mental states while being neutral about their sort of ontological character right so um so it could be then that if you do some you know neuroscientific investigation and you find hey there are these you know states in the brain that are they seem to bear the same relations to one another that these mental states that we were talking about bear to one another uh you know lewis says well then because this the sentence says that the mental states really are just the things that uniquely realize this structure of relations well you've just empirically discovered the mental states um so it's a really cool way of theoretically bridging um something like psychology intuitive folk psychology and and neuroscience and lewis's point was once you have this structure if you could extract this structure from folk psychology that's implicit in it and if you made the empirical discovery there'd be no further bells or whistles whistles needed to say that there's an identity between them because that's just what the sentence says is that the mental states are whatever things satisfy this um description very interesting so dean then stephen then adam yeah so so alex i mean as it's as it's written out here it's it's really stable and i'm i'm really curious because i really don't know this do different time intervals like really really short time intervals or longer time intervals if i've got a day to go down the stairs versus i've got one second to decide whether the 96 inch drop is less harmful to me than the lion that's chasing me do time intervals affect the stability of this right yeah so sorry i neglected that part of your comment earlier um i think no is the short answer i think that this is meant to be a static a

static sort of description of a implicit theory right which which will include statements about short time scales and long time skills so i would say that that anything about time scales in theory should be captured by the folk psychology that you embody or that you have in mind at the moment and that might change from one moment to the next also like this doesn't need to be a temporary static structure so maybe that speaks more your point but but i think the possibility of of spelling this structure out even for one instant is the is the is the lever that lewis uses to to draw his philosophical conclusions okay thanks thanks steven and then adam this question about prediction and time i think like you just saying there may be a sort of time scale at which things are just flowing like what sort of speed is it that things are just doing their thing i.e cells are doing their processes but what dean was talking about was is longer time scales than that and i think like you say there's something different there that's a more everything beyond a certain point has to be imagined i suppose anything that's not possible to do variational free energy on you've got to do expected free energy and to do expected free energy you basically have to imagine some a big us imagine in a different way to just our limited sense of imagination like our body has to imagine at some level the expected free energy basically what expected free energy is so i think that's um that then does bring in some interesting ideas around folk psychology would you say that those mental states can ultimately be action and sensory states that are being predicted so the mental states are basically the recapitulation or expectation on sensory states and resulting beliefs on action states so they they could be seen as mental states but actually there is a more distributed realization of that i'll be interested if that that sort of is something that could hold yeah i mean i think that the spirit of lewis's proposal is also functionalist right so so the the basic idea functionalism of course is that the mental states in the philosophy of mind is that mental states can be defined in terms of their relations to inputs and outputs and also to each other and it's i guess it's the to each other bit that distinguishes it from behaviorism so um but yeah i think the these implicit definitions that lewis talks about would rely heavily it would depend heavily on things like um when i believe that there's a cat present i expect certain visual appearances to occur um and i'm disposed to act in certain ways and maybe you want to catch the the i i tend to want to cache the actions and the sensory stuff out in terms of proximal stimuli and like motor effector states rather than distal things and extended things that's just i i don't know that's my my my um i don't know if it's a bias or what that's just the way i i tend to think of these things but um definitely i i guess i would say that the the relations between the mental states though are just as important as the relation to uh the active states from the sensory states um when it comes to

this kind of um this kind of formalism um i feel like i missed an opportunity to it was a really interesting comment actually i i don't know if i did it justice but i'm still thinking about it can i just just add one little things like just before yeah okay very quickly but one thing that comes to mind that i i think might connect what you're saying um you were talking about this and and and so there seems to be something there about um the combination of ands um and what's you know so for instance and there was a dog and there was a woman with the dog and were in the park and the dog was wagging its tail and then there was blood on its teeth when it smiled and i'm like suddenly that changed it right so the triangulation of the ands flips once you suddenly the dog bears its teeth and you see blood on them so that links into what they do in qualitative research where they often triangulate to get something which has got trustworthiness rather than accuracy um and i think there's something interesting there about um what you're saying in terms of that my knowledge of that field of linguistics is but i if i can bring it back into embodied work i find it easier which is what i'm trying to do but this and and and and that's that has got some practical applications i think in um in a number of areas definitely yep thanks adam um if you can return yep okay go for it adam hey making coffee so um eventually i'd like to get to um free will and actually uh loop around a little bit to actually depression as a case with respect to that but um before we move on from ramsey's sentences um uh not to belabor it but so it is the idea that so you strip away the specific entities being signified you look at the syntax and then you're saying it can the same syntax apply in another system and then you have the mapping that this would be the map and so that would be that the basic idea essentially yeah if you take syntax in a you know in a slightly broad sense then totally yeah okay so i guess i'd be wondering like um like in terms of like active inferential modeling would it be something like um like with like the graphical brain like you have like this forney factor graph where you have like the continuous regime down below but then you have this like discrete regime up top and that maybe something in like the factor graph portion you can look for like some sort of like uh homomorphism or isomorphism there or like if it's the brain you would do some sort of like you would look at ensembles and like describe them as like attractors moving along some sort of manifold and then like when you core screen those or throw a blanket around them or reify them in some ways if the syntax of their interrelations matches people's uh uh subjective reports of their then you would be able to actually potentially identify like physical sub physical and computational substrates of consciousness is that i mean yeah i have a bridge too far or like that well there we're going that's i think that's where where we're going except uh i never made the bridge all the way to

consciousness because that's one thing that i i reserve the right to not decide about i think your work on this is really cool actually i kind of suspect that like most of the main theories of consciousness are kind of overlap a bit and there's a core that's you know that's sort of like uh getting at the way things are which i think you've written about but um so so i want to just bracket the consciousness thing for a second although maybe that totally ruins it for you i don't know but like i think yeah but that's that's exactly the idea like i think you just expressed it very well that um if you look with yeah you have to squint you have to do some coarse graining but if you look at the dynamics of the physical system uh you'll be able to discover in the ideal case some of the same sort of syntactic relations that you would get from the psychological perspective and i think that's also very constant with the way you know connectionist modelers describe their work sometimes so like you know you think a connection is a simple you know neural network made up simple processing units uh you can derive that architecture just by looking at the brain and sort of very much simplifying things or you derive the same graphs in some case just by looking at cognitive processes and so you know it's the idea that you have two perspectives on one thing if you have a graph isomorphism then that is what was being hinted at with it being sufficient to find that there's relationships amongst mental states like i always feel like i feel this way before this other feeling and then if you found that there was some brain state that always preceded is that the end point or is there still more to explain in that context let's actually get to another topic that we highlighted last week which was the sleep wake example so i'm here on slide 40 and we're considering a system whose analysis is tractable at least according to alex and we have an agent whose behavior as far as sleep wake is concerned is defined by equation 3.1 so could we walk through what is happening in this simulation like what is the agent doing what does the agent consist of and then what does the equation show and then how does that relate to what we're showing in this paper sure so i think i think this is another aspect of the paper that um that was uh maybe seemed to be a making a deeper point than it was but i'll just talk about this yeah briefly so this was actually just meant to be uh sort of a slight modification of just like a helmholtz machine running the wake sleep algorithm so i don't know if i need to set that up blue talked about that a little bit in the point zero um but the idea was if we just have a generic let's we have a neural network that has a generative model that's sort encoded in its top-down connections and a approximate recognition model in the bottom up connections and so so one way of doing this is to run the wake sleep algorithm where you basically generate uh states of the network top down like fantasies right functionally speaking and then you adjust the bottom

up weights so that they're likely to produce those fantasies and then you alternate this with a so that's the sleep cycle you alternate this with a wake cycle where you use it input of the kind of sensory data you want the model to learn to generate and you do a bottom-up pass and then adjust the the top down generative connection so that they're more likely to produce those states that are induced by input okay so um so all i was doing here was saying uh look if we if we suppose just make the assumption that this is a real system we're talking about it's ridiculously unrealistic right so this that's why i said it's uh it's it's something that's tractable to analyze but it's also quite a far cry from you know anything biologically possible but it's not also it's also not completely off the mark right so i think there's a reason that the helmholtz machine was kind of like a seminal moment in the evolution towards what we have now um because it got a lot of the coarse grained um sort of conceptual structure really down that i don't know if it had been pulled together before in this form anyway so let's suppose that the wake and sleep cycles of that machine uh occur with equal probability then we can talk about probabilities of certain states occurring in the network just period right whether it's during a sleep or a wake cycle um and so if you just grant that if you if you ignore the fact that this is a really only it ends up being an approximation to an approximation right so it's it's um it's running it's uh wake sleep ends up being sort of an approximation to the em algorithm which is itself not quite full variational base so anyway lots of approximations but if you ignore that for the moment i just wanted to make the point that you could talk you could talk about the you can thoroughly completely characterize the possible probabilities of various states of the and so you could use that to define this this r here is just the probability of a certain state of of the system being in a certain state now here the s_i is referred to like the states of the individual neurons but that's related in a simple way to the overall state of the network um so you can talk about the probability of the system being in one of these states um regardless of whether it's running awake or a sleep cycle and then that gives you a if you have a probabilistic description of the states of the whole system uh then you could use that as as um to define the the relevant terms in the thermodynamic uh free energy so the entropy and the energy and such so that's the basic idea but the the part that you that you were wondering about like what is this um system simulating what does it perceive and such that's all that's this example is completely agnostic as to any of those details so once again this is highly highly abstract and i apologize no it's cool and what do you think this model would allow us to continue developing towards or what does it show for the argument that you put forth as far as the psychophysical identity thesis right well i i don't know that it shows anything in itself i

think what was what it was trying to do was was was write down a system in which the target would be clear so like the you want to be the case that this r distribution um so so let me put this way so the r distribution because it's it's essentially made up of or it's it's factored into these wake and sleep cycles right those have direct um interpretations in terms of variational free energy and inference um and yet we we have a well-defined distribution over the states of the that we could use to describe its physics so i just wanted to show that there's one simple case where where we can we can write down a distribution that bridges these two descriptions and then uh then in the subsequent paragraphs i try to i try to argue that we can generalize this beyond equilibrium and that's the that's that's the very tricky bit and i don't claim to have shown it by any means i'm just again i was trying to remove obvious obstacles to this thesis being true awesome so adam with a raised hand and then blue or if anybody else wants to go for it um in terms of um wake sleep uh there could be some uh potentially biologically realistic implementations of something like that um o'reilly and um has a paper uh deep predictive learning where he describes a model of the thermocourtical system doing predictive processing that's a very weak slate a wake sleep-esque i'd be um i'll send that to be curious enough if you like if you thought that was like a good like psychophysical identity mapping there the other uh thing would be um so if if we think of experience as being the ramsification of experience with the mapping would happen at the level of um then and if the predictions are the uh sleep part um is there a sense that experience is always like kind of paradoxically it's always the dream like the awake part is the stuff that's just updating it but the actual thing generating the experience is the dream part like correct or yeah no i yeah so i i have an unpublished paper on this too on the representational division of labor like so people like in predictive processing literature say things like yeah we experience our expectations of the world or whatever but i'm not i'm not so clear that it's that way i don't know um i guess if you think that the predictions are the carriers of the experience or the neural correlates of the experience or something then then that would be the case and the wake cycle would do nothing but i think it's i don't i don't see the reason to embrace that that that interpretation of of what's going on here um because it's awesome no i mean i'm all about no no i'm all i'm all about the idea that that we experience sort of virtual models in a way right that's that's cool i just i i don't interpret that in terms of like it's just the it's just the top-down driving signal that leads to experience although maybe i'm not really against that either and i can see why you'd want to say that um anyway i just don't think that there's any inference from like the helmholtz machine to that like because and i would love to look at this paper you mentioned by the way

on like biologically plausible versions of this because i think there's a lot that's right about the only the reason i say it's so impossible is just that it doesn't have any role for top-down modulation of bottom-up signals during perception which was by design so that it would work quickly um but that's the main limitation that concerns me um cool thank you alex steven with a raised hand and then anyone else and also anybody watching live you're more than welcome to write a question in a live chat so stephen go for it just uh just to help me clarify so this being an attractable pro analysis so the key the key point is that by going to stochastic it shows in principle how recognition and generative densities can be used together is that the kind of the point that this shows i just because i'm struggling i think i'm just i'm not so good with the formulas and to exactly get it but i think the general point you're making is you're going to stochastic algorithms and therefore you've got this potential for the recognition and generative densities which is basically sensory and accurate potentially to be reconciled and shown even a very simple case would that be a correct reason a bit i i think so i i think definitely that's um it's it's in large part right so that the idea that we're dealing with a stochastic system is kind of important because yes i want to claim that or i want to advance the suggestion the possibility that probabilities are encoded by probabilities so then you could have an actual identity whereas if you're just encoding like if you're using like i don't know the um a scheme that's more like what carl firsten talks about often where you have sort of neuronal states that are encoding the sufficient statistics of distribution then i don't think you could get an actual identity because you just you'd have a clear representation that has a certain um almost convention built in of how things are represented whereas i'm arguing for the possibility of a much more direct encoding and that does depend on this being stochastic but the other part of this that i forgot to mention that sort of i think maybe crystallizes point this is trying to make is that um if if we're at equilibrium with respect to the network if we're at the um helmholtz machine's equilibrium distribution then um in that case if you do take rs describing the physical probabilities of the system that you would use to specify the thermodynamic properties uh then you could show that in that case at least the variational free energy and the thermodynamic free energy would be the same and then generalizing to non-equilibrium of course is the hard part but um that's that's that's more concretely what i was trying to show there um yeah thanks so i'm gonna go to a question from the chat and then we'll go to adam so cambridge breaths asks can this model describe or map the transition between focused attention and mind wandering grateful for any elaboration thanks everybody um anyone want to take that i i don't know i don't know yeah i don't know just maybe the dream versus the you

know sleeping versus the waking but then even within the waking state we have a wandering phase versus a very tightly focused phase so there's sort of what is if that reminds you of anything i mean i'd say if we're talking about this model talking about the helmholtz machine or this like slight elaboration on the helmholtz machine then i don't think it would do anything that sophisticated just because like so as adam says there are biologically possible ways of implementing this kind of thing but like anything is fine-grained as a shift from attention to mind wandering i think is probably going to be left out of a model that's at this uh level of sort of abstraction and approximation that's my thought thanks so adam then blue and then anyone else with a raised hand or a question in the chat adam i'm not sure i can address the mind wandering exactly with psychophysical identity but in terms of the mountain climbing analogy i'm wondering if it's like mind wandering is kind of like uh exploratory hill climbing with a little bit of quantum tunneling something like that but uh that's a that's a i mean what would you what you think actually it might loop around to the brain in an interesting sense um and that um there's like if we think of like lower levels of like a cortical hierarchy as doing something more like um a variational inference uh via like continuous message passing maybe with like some discrete updating there too that seems to be like one kind of implicit representational regime there might be a sense in which you're getting like probabilities in terms of like the attractor dynamics along this hierarchy corresponding to like probabilities of like a hierarchy of nested events in the world and like a deep temporal hierarchy and the deep temporal hierarchy of like the latent causes of the world uh but there seems to be like potential like as we like move inwards like another level where these get um re-represented into more like constant like when you're into like deep association cortex it might be a different game and then when you're getting like coupling with the the campbell system uh now it seems like you're getting something almost like like explicit syntax with like discrete semantics because you can map like there's these pointers going from like the relationship among these like you call like bump attractors the hippocampus system or like these place cells but they can be used to like contextualize the whole overall uh system within some sort of uh structured syntax um i'm i don't know like would do you need something like that to have the probabilities where for um is that what you're looking for or could there be more like this like inactive hierarchy of uh like a hierarchical boltzmann machine that has a nested and typical cause there and that those are probably like yeah both do it or would only like the re-representation yeah i don't know i think a lot about this this issue of re-representation and how much you know if it happens and what it's doing and stuff but um i think uh i guess i need to understand more and this is totally

just my ignorance about the kind of models you're describing with hippocampus and such like um i tend to think of um i tend to think of semantics in the brain just in terms of like vector space models um and so uh i'm just curious i don't know if we could talk about it now for a bit maybe i don't know it's up to daniel and others but like um what is the explicit syntax in this case it sounds something like something that i that would be friendly to to what i'm trying to do um so i guess like uh so this explicit syntax it's like still heavily researched there seems to be kind of like a gold rush among different like ai cop companies and like cognor labs of trying to figure out hippocampal system like deepmind this is kind of like what started them it was like kasabi's like making discoveries about this but the idea would be um that you have graph structure that would be that originally would have evolved for just locations in like uh meat space or physical space yeah but then got repurposed for any kind of space and so this gives you like structured state spaces whether you're navigating through the world or you're navigating through like a spatial topology right this would basically the content would be these um and so it would be at the top of the cortical heterokey um and it would have pointers that could be unpacked in terms of like these more like modes of an active engagement there might be like another like intermediate level could be associated with consciousness where you're getting like um more concentrated attractor dynamics with like potentially a quasi-topographic relationship to events in the world and so the principles there um uh graph neural networks might be relevant there where basically you get orders of magnitude greater representational efficiency by uh actually making the geometry of the system over which you're doing uh the deep learning resemble the thing that's being uh modeled so like um there's like graph grid neural networks that can be used for like spatial modeling or there's like graph mesh neural uh modeling of like someone's po like a a pose of a thing or an object and so in theory some things like this might form at the intermediate level and these might be the ones they're most heavily coupling with this hippocampus system which is giving you this syntax of state transitions right with like structured composition that's really cool can i comment on that for a minute yeah um alex go for it and then blue okay um yeah so there's a lot there that's that's exactly along the lines of what i've been thinking about this i've written a bit about these like conceptual sort of grid cell type things uh that it seems like there's a lot of evidence for that it's really awesome i know like memento is doing some research that's related to this stuff um so that's okay if that's what you mean then i'm totally on board like i wasn't sure if you meant that or something more language like with like discrete tokens with other parts but i mean that might be encompassed by this as well but it seems like uh people are finding also like

encodings of like the hierarchical uh structure of like the grammar of language is also encoded uh it's found to have a correspondence uh right with like the structure of the like they said the relationships of these bump attractors that's amazing yeah i was going to mention also so you meant you gave all these examples of graph of neural networks that are sort of isomorphic with the domain that they're they represent um i was going to mention recursive neural networks used for language processing as well the the the network topology matches the syntax tree or something like that so i that's very that's very much along the lines of where this is going thanks alex so blue and then dean afterwards so i want to kind of loop back around to what adam said about dreaming because he totally like took the words out of my mouth but maybe um also touch on the question that was asked by the person in chat about mental wandering so is it possible so so alex your um you know objection to the dreaming and the physicality was that there's no evidence for top down control during the wake cycle is that correct is that what i heard you say um sorry no i actually meant to say kind of the opposite i meant to say that um that the problem with the weight with the helmholtz machine as a model is that it doesn't it doesn't model top-down effects during waking perception right and that's what i interpreted so so okay so if it doesn't if it doesn't model weak uh top-down effects during the wake cycle um is it possible that maybe there are like overlapping helmholtz functions happening right like so you know your mind wanders off or you're not paying attention and like maybe you're consolidating like you're you've got like one set sleeping one set waking and there's maybe like an overlapping construct of this like wake sleep learning cycle happening like while you're out mental wandering i'm consolidating you know the information from you know what i learned at breakfast or or something like that yeah yeah i mean i think there's i i think there are there are like like this model is adjacent to like many really interesting possible variations right and slight elaborations so like what you're talking about sounds to me like in that vein like i saw an amazing git repository of just like helmholtz machine variations right where they just they just tweak certain parameters like let's see what happens if we propagate a few steps down during the wake cycle as well you know things like um so i think i don't know so i don't know if i fully grasp like exactly what you had in mind but it sounds pretty good and thanks for i mean you said go ahead oh alex go oh no i was gonna say thanks for also trying to address the comment and because i felt like i didn't wasn't able to answer it very well nice and there's definitely wandering in dreams and how is that different than the wake wandering and could that get at what the difference is between sleep and wake and could this model maybe give us a bit of a wedge to enter into that discussion so

dean and then anyone else with a raised hand so alex back back to analogizing so i'm reading your paper and i can read something and i go oh my goodness look there's the world's biggest nickel and then i continue reading and then oh look there's the world's largest corn cob and then i have a chance to have a conversation with you and you're like saying to me well let's bring this back into some sort of proportionality so explain to me what is 2 carat deep diamond that you were trying to allow readers of your paper to get to because i know it's i know i blew it out of proportion because when i first read it it was a big deal but maybe you can help me bring it back into some sort of scale as to what what was your deep hope here yeah it's interesting no i mean i didn't i i don't i also don't mean to suggest that you got it wrong or anything with i think the proportion you're finding interpretations that are consistent with what i said you know it's not like it's not there but um and it's funny that if i'm trying to if it seems like i'm trying to put ad to put things in proportion or ground it in any way it's kind of funny because i feel like other others are trying to are trying to do that like you know relate this to embodiment and activism and things much more concrete and i i keep trying to keep it abstract but um but anyway the main i don't know what the hope was here i think yeah i'm not sure i i think i think i just really wanted to point to the possib it was really about this identity thesis right which i'm not even sure that priyary i agree with by the way like i'm not sure that identity theory is the way to go in understanding the mind's relation to the brain i'm kind of like not really a reductionist but um i wanted to i it seemed like there was a possibility of of uh articulating a like quantitative version of that right using the tools that we've developed over the past several you know a couple decades that lewis didn't have access to um and i just really i i just love david lewis as a philosopher i think he had some amazing ideas and i i wanted to uh and it just seemed to it's not so much about that i wanted to to show anything in particular it just seemed to me that there was a natural convergence here that that could be spelled out so um that's kind of what i was trying to do thank you so steven with a raised hand and then anyone else who raises their hand and also in the last half hour or so anyone who wants to ask a question in the live chat so stephen then adam um i was just sort of tying in a bit with what was also said by adam uh around the the hippocampus so you've got the place cells in the hippocampus which gives the there's some way to bring the environment in potentially in this semantic i like i've not heard it thought about as semantics but a semantics of place if that makes sense so now we've got our environment external states potentially being brought in in an attractive way you've got action states could be the grid cells with the whatness of what's out there and the sensory states is this lower level flux that's coming into the bottom of the

brain and then the the generative model would be the the the higher level kind of well i'll be interested what the generative model is in all of that that if that makes sense if if we were to play in that in that sandpit it would it would give that way to bridge that i'm a big fan of the hippocampus let's just uh but i i'd love to see how it all fits in with some of these pieces i'm just curious around if that if that kind of uh role for these grid and place cells could be quite useful to to do this tractability between higher and lower level states here alex if you have any thoughts there otherwise i'll go to a question from the chat and then adam um yeah i mean i guess just really briefly it seems to me that if we if we are using these these place cells in in this sort of we've leveraged this thing that was originally meant for physical spatial navigation and we've transposed it to like conceptual spatial navigation essentially that's really cool i could see it being the sort of the thing that conditions expectations for the sensory motor stuff um and sort of it would just sort of define the generative model i guess the dynamics at that highest level would be the thing that sort of sets the the set points for things but i think adam might have more to say so just to the question which is also for adam says uh thanks adam did mention that the transition process probably involves quantum tunneling may i ask if he thinks the same for the transition from mind meandering back to focused attention so adam feel free to take up that question as well as anything else you'd like to add and then anyone can raise their hand oh boy um i mean i guess like um there's a sense in which like one of the quantum tunneling events i guess i would have in mind so like there'll be like these uh regimes of sense making where the hippocampal system will lay down a given tiling of the hippocampal system and collaboration the anterior cortex of the grid cells will lay down this tiling of some domain within which you're doing this modeling with where that you can get basically the relationship among these bump attractors you can get this like structure the syntax and through the pointers the kind of synaptics uh and also with its themselves having a semantics of space maybe um just in the relation but the so i guess the quantum tunnel like so so but then if you get enough prediction error from the overall system this seems to trigger these resetting events where you'll break up the frame and then you'll do a new tiling and it's kind of like a re-grip on the active inference it's like a new kind of engagement a new set of policies that could fit within this different uh conceptualization of the safe space you're working with and so um these events of remapping i guess you can maybe think of them as like the quantum tunneling it's like you're just you're not doing i don't know if this works but it's like they're called like um some people come like levy flights like an animal is like foraging in a patch and then it's like and i'll just like jump to a greater patch once it's like not getting enough and so it'd be like

you're kind of like imagining in this place and you're simulating this place and then you're like okay a little bit bored not quite the thing and then now you're imagining something else then you're imagining something else i guess you could think of those as like quantum tunneling like you're not just like working within one regime of simulation just fixing that thing you're moving from here to here to here but uh the question i had when i raised my hand was um so there's a sense in which you might think of like predictive coding style models of cortex i know it's like more complicated than predictive coding but there's a sense in which something like that is true probably true because if you do a primarily suppressive regime you end up inducing sparsity and if the events that are in the world are clocking slower than your internal updates you come out ahead in terms of like i mean that's why they did it for video coding it's just more efficient and so i'm wondering is is that enough to give like a psychophysical identity mapping um in terms of like there's a sense in um you will minimize predictive coding mechanisms so they say we can go with just those um they should minimize uh they should minimize activity yeah um yeah that's that's actually kind of where i'm coming from with this paper um in a to a large extent i was thinking about yeah predictive coding as a way of understanding neural networks um so i think although this perspective can uh can encompass you know active inference and such i think that's sort of the core of it is that brain is literally trying to yeah to um right sparsity as you said right induced sparsity minimize activity only pass forward signals that need to be passed forward and that means that you're sort of quelling activity that doesn't get passed forward so i think i don't know if that's strong enough to get you an identity but i think it's uh it's consistent with the identity idea and it's um i think it's a large part of the inspiration for me awesome stephen with a raised hand and then anyone else yeah if you could just speak to how you see that predictive coding predictive processing be more or less useful at certain times as the sort of the the the discourse and when maybe active inference is not as needed at certain times you know thinking in applied context but um i'll just be curious yeah um it seems it seems to me that if you like that if you want a like the the the main advantage of active inference practically speaking to me seems to be like modeling uh like explicitly how how policies actions are chosen like decision making right based on um how you expect your actions to change the way states evolve so i think that that happens implicitly in a well-trained predictive coding architecture right i and i i see i don't see any fundamental theoretical advantage to active inference there in terms of explaining how things work on like a in principle level but that said it's really hard to like just train a recurrent neural network or whatever to to do what you want to do and then interpret its states and so on so if you have an

idea of a generative model like of what the agent's generative model should look like then it makes much more sense to write it down a priori and you know try to build an active inference model than to just learn it from data um so that's as close to maybe a applied setting as i could get but i hope that sort of speaks to it yeah let me let me try to get another view on that from you alex where would you put active inference in relationship to supervised and unsupervised learning for those who might be less familiar with those topics when you're talking about learning from the data versus a generative model where does active inference fit into that yeah that's a great question um it's it's a bit it's a bit tricky because there's this issue of preference learning that's kind of not settled in the you know like as far as uh how how much of the future of of real uh biological agents uh learning do we think that is but like um so to me the most natural way of thinking of of unsupervised learning happening in active infants is that you've got sort of a um you've got a phenotype right some some aspects of which are going to be immutable over relevant time scales but some aspects of which will change right so we're talking about synaptic weights you could think of that as an extended part of your phenotype and that will change over you know time scales of perceptual learning and and during the life organism's lifetime and such so like um that that is i think is neatly you can categorize that as a form of unsupervised learning um but that's also not really uh again it's not it's not there are people who are active in sort of practitioners who don't really uh think that preference learning is a thing that happens or um yeah it's not clear what role it plays um but but but overall i mean uh well there's also there's also gotta be a role for um something like uh supervised learning seems to me the closest thing that we have to that in real systems is is like uh reinforcement in a way so like uh and that's certainly gotta play a role that's complementary to unsupervised learning but i think i don't know this this answer is kind of i'm sorry uh it's uh i'm happy to to to take it up more i'm just not sure what like i don't know how to how to how to tease apart the you know the the parts of active inference that should be understood in terms of unsupervised learning versus just like learning in general and how much of it i i suspect a lot of learning in general has got to be unsupervised and i think active inference is sort of uh naturally interpreted it's predictive coding certainly is is is an unsupervised learning paradigm and to the extent that active influence reduces to that it is too great thank you and these are definitely all topics we're always trying to think clearer about and communicate clearer about so stephen i think this is a useful conversation i know it's a bit off it's pretty on the spot a little bit but i think as a group it's quite useful this question of um when a generative model comes into play so with reinforcement learning i suppose there's a generative model in relation to like

an induction an inductive reasoning in terms of the goal like you're trying to narrow the gap between the target that you're reinforcing the reward and reducing the gap whereas there's this abductive potential in active inference to and i wonder whether in the in the population or the population of the brain's attention i'm sort of wondering if place cells and the sort of more granular parts of the lower levels of the neocortex or i'm not sure if that's the right term are basically that kind of um more unsupervised very unsupervised very abductive kind of potentially ways of engaging and then these grid cells as they get more populated with what things are become kind of more top down maybe more let us go back into say a more reinforcement learning or supervised learning because like and then you you don't forage as much so to speak you uh you just go straight to the point which probably i should do now and let someone else thanks steven um alex if you want to add something otherwise we'll go to adam um yeah no just to say i mean that kind of makes sense to me i think i think you you need i think you need unsupervised learning for representation learning of whatever isn't innate right and then once you have that framework then then you can start to think about like reward like meaningful you know interpretations of sensory stimuli that you could associate with rewards so that's all saved for now cool adam um a quick uh com a question but a quick comment on what stephen said um i mean interestingly about hippocampal system is there does seem to be like like very contrastive thing to it um in terms of like comparing between current states and then imagine states and like at different like duty cycles of theta you'll be seeing there seems to be this comparison operation going on so it could be very much like this higher level highest level supervisor in this like uh matryoshka dolls of like quasi-homuncular things that we're thinking of um but um the the question i i was wondering wait there's people who don't believe preference learning is a thing oh i shouldn't i shouldn't have put anyone on the spot it's not that it's necessary they don't believe it's a thing it's that it's not clear that we need to invoke it to explain behavior behavioral things in at least in in many cases um so i feel like i'm gonna i'm gonna be speaking for for people who are colleagues of mine who i don't wanna i don't wanna misrepresent their views so i'll shut up about it but there's definitely controversy about the extent of which it happens i suppose i mean i i'm wondering i guess like to what ex like so like there's things like uh like metal learning uh principles where like you have like you're figuring out like new ways of approaching policies so you're like of realizing goals and you're building up these like policies like an exploratory fashion as part of your phenotypic plasticity and um but i guess there is a sense in which like if we're like for subscribing to the free energy principle you can only prefer one thing and that thing never

changes and everything else is just instrumental it just you could just only prefer to be basically agent smith for the entire world you could like my pattern me me me um right but you know ultimately this should be like we in terms of like evolutionary favorable equilibria but it i guess there is a sense in which that never changes yeah i mean i just so i say one thing to like to try to make defend the people that i i've been talking about um but i say preference is controversy about preference learning i mean there's clearly something that that looks like preference learning right it's just a question of whether you're really learning preferences or you're just learning new associations with with the things that you prefer and the complete those two processes are conceptually distinct they might lead to similar results but uh yeah but the point you're raising is interesting that is that really your preference is like you right literally your phenotype so well yeah i have a question for you yeah that's exactly where i was going to go could there not be a cultural scaffolding that's developed over ecological and evolutionary time that values diversity of perspective and so yes it is true that at different scales some type of active inference could be happening but i don't think that agent smith would be the only attractor at the bottom of that bull it'd be more like a uh maybe a plate with a lot of different wells a lot of different kinds of food that could be converged upon why would it have to be converging on one specific um especially hollywood represented version of what efficiency is just a thought but yeah i don't think agent smith would win like villains usually lose you know steven well this this could be though when you get into that non-equilibrium equilibrium dynamic is like once you get a dictator who establishes an equilibrium that's an energy equilibrium which is that's going to be more powerful than the more subtle organic non-equilibrium states it just dominates at times and if there is made in equivalence in some near or distant future between actual energy usage and governance for example some sort of crypto system with energy and voting and information and finances all being mixed together it'll be a enormous nexus of power and maybe we'll be able to use these kinds of frameworks describe them stephen and also going the other way then if you this is my belief this is without is i i strongly believe that to create community engagement in meaningful ways you have to create the the container um for those more subtle non-equilibrium states to have a chance to breathe um this this woman ariane munchkin she's a she's a theater director in france she talks about don't crush the butterfly she will tell the actors don't crush the butterfly the butterfly is like the idea so don't grab hold of it because you'll crush it so it's quite a nice metaphor you have to hold it lightly and allow it to and there's there's a certain because equilibrium dynamics or someone who comes in and basically starts shouting at people will always over

dominate someone who's trying to do some let's all come together and find each other's plurality so anyway that's that sort of dynamic sort of ties in with what you're saying cool so i flipped to 49 where we have our usual set of just closing discussions so in these last 10 minutes we'll just be remarking on what we thought was interesting or important from the conversation what would we like to pursue next in terms of our own work or alex in terms of your research i'm sure we're also all curious as well about how you apply these ideas in terms of your you know whatever non-proprietary applications you think that this bears on if none then that's its own interesting claim or maybe there is something that this helps us do in terms of practice um what are the next steps here maybe alex first and then anyone else can give a thought uh sure so i mean i could address these questions on the slide or yeah let's let's yeah let's hear your question your answers to these questions on the slide so i mean this also speaks to what you just asked but i think really i think that this is just a way of thinking about things um that the main thing i think it enables is just allowing you to in to infer from what you know about cognition or the cognitive states of some system including yourself to physical states right if it's true then for example yeah you can say well i'm depressed there's not a lot going on um you know i'm i'm uh i'm ruminating a lot uh you can infer that there's something physically happening that's mirrors that process and you can take either a physical or a psychological approach to solving any in your body right in theory now there's a lot of working out to be done of how that would work in specific cases but that's the kind of like basically just sort of smoothing the path for inference from physical to psychological states i think is uh is the main thing that i think is is practical about the practically important about this um and i'm still curious about how to spell that out um for specific types of states uh like like depression desires things like that and um i'm also curious about whether any of it is true at all or whether it's just complete nonsense big questions adam and then anyone else who wants to give any thought or address one of the questions on the slide if there's complete nonsense i'm not sure what would make sense i'm not even sure that what the sense would mean um yeah uh later i would really want to talk to you about like the degrees of freedom and like volition issue and and like different ways that could play out and also i want to drag you into consciousness because you you are precise my friend and you're helping me think better so cool thanks adam happy too cool um if anyone else has a thought otherwise we can sort of look back oh yeah steven go for it yeah i just will say thank you for bringing the sort of this foundational build-up into the um thermodynamics and i'd be interested to yeah maybe sort of see if there's any i i did physical chemistry um as um so i've got some background but i i i'm curious to see if

there's any ways to um bring in some elements of of sort of gibbs free energy and uh into some of this so i'll be glad to chat about that at some time yeah blue so when you guys are done talking about degrees of freedom and agency you have to come back to the chat because i want to hear all about this i have some thoughts that have been cooking about agency related to active inference other ideas also cool and maybe dean i'd like to ask where does the dot two get us what do we do at the end of the dot two we took that step to the dot zero with yourself and blue and then we take the dot one and dot two steps ideally with the author especially when it's an awesome author like alex who's engaged with the material before the dot one and between the two what's our next step you know on that staircase what do we do after the dot two how do we make the best of this well this is a sample of one so but for me it's uh so how do you remain stable and dynamic at the same time how do you move through these gradients whether it's ascending or descending and also keep in in the front of your mind how you're going to keep your grip if you're ascending or maintain your balance when you're descending and for me that's what alex's paper essentially pointed out there is a there's a physical piece to that and there's a statistical piece to that and i don't want to become a statistic of another old person who falls down the stairs so how do i pull all of that together and have it make sense especially if i don't want to give somebody a training exercise if i want them to be able to model how to get down that descent or up that ascent and i can't give them a map so what do i what do i give them what do i replace that with that's what alex's paper gave to me gave me a little bit of confidence awesome one more staircase thought and coming back as well to the up and down hill so when you're going up the staircase you'd grab onto a certain spot on the rail and then that helps you ratchet up and then when you're going down the staircase you're unlikely to grab on tight to a rail you kind of want to loosely have your hand sliding down the rails so that you can ratchet if you needed to and stabilize but if you ratchet while you're going down then it's gonna prevent you from continuing this sort of natural descent so it just makes me think about different modes in understanding especially because as we talked about with the mountain climber it's like going up and in their own way if the top is your goal they're like both going downhill so when in our process do we want to grip the rail lightly just like steven said not crush the butterfly just slide along and just go with the flow knowing that we can stabilize when we need to and when do we really want to grip and ratchet because we're working against one energy gradient but we're working towards our goals when we do that so i hope all these mixed metaphors come together for the listeners in an enjoyable way and alex we really appreciate your engagement with actinflat here and everybody else awesome

conversations yeah thanks thanks uh blue dean daniel steven adam uh people who were here last week also um i'm i'm happy to be put on the spot i'm glad this was of service to people i think it's been a really interesting discussion and um thanks for thanks for having me thanks for for all all that you do great well everyone's always welcome and we'll see you all in another stream bye